

Question on notice no. 559

Portfolio question number: DHDA SQ25-001584

Supplementary Budget estimates

Community Affairs Committee, Health, Disability and Ageing Portfolio

Senator Alexander Antic: asked the Department of Health, Disability and Ageing on 17 October 2025—

During the Senate Estimates on Thursday 8 October 2025 TGA advised that since February 2021 the TGA has investigated 148 safety signals for Covid-19 vaccines resulting in 57 updates to product information and consumer medicine information documents.

17. What are the 148 safety signals that have been investigated?

18. What are the 57 updates to product information and consumer medicine information documents?

19. How has the TGA complied with the COVID-19 Vaccine Safety Monitoring Plan?

20. Could the TGA provide documentary evidence demonstrating how each of the five key strategies of the COVID-19 Vaccine Safety Monitoring Plan was put into practice and the documented outputs of that plan including but not limited to:- Documents evidencing the TGA's implementation of the Plan's stated aims and key objectives being:

- a. timely collection and management of reports of Covid-19 vaccine adverse events following immunisation
- b. timely detection and investigation of Covid-19 vaccine safety signals
- c. timely action to address any Covid-19 vaccine safety concerns
- d. timely communications to inform the public of emerging Covid-19 vaccine safety information and to support public confidence in vaccines
- e. close collaboration and coordination of effort with other vaccine safety stakeholder groups.

Documents evidencing the TGA's implementation of the Plan's five key strategies including the TGA's implementation of each of the strategies, objectives and outputs referenced in the key strategies of the Plan as follows:

- a. Enhanced reporting of Adverse Events Following Immunisation.
- b. Enhanced safety signal detection and investigation
- c. Actions in response to safety concerns
- d. Communications
- e. Collaborations

Any revisions of the Plan.

Answer —

Please see attached answer.

Table 1 – List of COVID-19 vaccine safety signals investigated by the TGA

Vaccine	Adverse Event
Vaxzevria	Guillain-Barre syndrome
Vaxzevria	Toxic epidermal necrolysis
Vaxzevria	Capillary leak syndrome
Vaxzevria	Venous thromboembolism
COVID-19 vaccines	Herpes Zoster infection
Comirnaty	Myocarditis/Pericarditis
Vaxzevria	Acute disseminated encephalomyelitis, encephalitis, and encephalopathy
Vaxzevria	Thrombocytopaenia including Immune thrombocytopaenia
COVID-19 vaccines	Hearing loss
Comirnaty	Appendicitis
COVID-19 vaccines	Trigeminal neuralgia
Vaxzevria	Guillain-Barre syndrome
Vaxzevria	Venous thromboembolism
Comirnaty	Bell's palsy
Vaxzevria	Bell's palsy
Comirnaty	Hypertension and Blood pressure increased
Vaxzevria	Blood glucose increased
Comirnaty	Lymphadenopathy
Vaxzevria	Guillain-Barre syndrome
Vaxzevria	Capillary leak syndrome
Comirnaty	Myocarditis/Pericarditis
Comirnaty	Myocarditis/Pericarditis
Vaxzevria	Capillary leak syndrome
Vaxzevria	Thrombocytopaenia including Immune thrombocytopaenia
Comirnaty	Risk of facial swelling in people with a history of injections with dermal fillers
Comirnaty	Myocarditis/Pericarditis
Comirnaty	Hyperthyroidism
Comirnaty	Cerebral venous sinus thrombosis
Comirnaty and Vaxzevria	Anosmia, Ageusia, Dysgeusia, and Parosmia
Comirnaty and Vaxzevria	Menstrual bleeding disorder
Vaxzevria	Giant cell arteritis
Vaxzevria	Thrombophlebitis (both superficial thrombophlebitis and thrombophlebitis)
Comirnaty	Acute pancreatitis
Vaxzevria	Bleeding events
Comirnaty	Bronchitis/lower respiratory tract infection
Comirnaty and Vaxzevria	Iritis/Uveitis
Comirnaty	Laryngitis, Tonsillitis, and Tonsillar inflammation
Comirnaty	Oral herpes and Mouth ulcers
Comirnaty	Chilblains
COVID-19 vaccines	Acute Kidney Injury
Comirnaty	Nephrolithiasis
Comirnaty	eosinophilic gastrointestinal disease
Vaxzevria	Venous thromboembolism

SQ25-001584 - ATTACHMENT A

Comirnaty and Vaxzevria	Cerebrovascular accident
COVID-19 vaccines	Acute mental state change
Comirnaty	Erythema multiforme
Comirnaty	Glomerulonephritis and Nephrotic syndrome
Comirnaty and Vaxzevria	Liver injury (including autoimmune hepatitis)
Vaxzevria	Balance disorder including vertigo and vestibular neuronitis
Vaxzevria	Acute myocardial infarction
Vaxzevria	Cholecystitis/Cholelithiasis
Comirnaty and Vaxzevria	Heart rate/rhythm disorders
Comirnaty	Cardiac failure
Comirnaty and Vaxzevria	Stress cardiomyopathy
Comirnaty and Vaxzevria	Vitreous detachment/floaters
Comirnaty	Anxiety related reaction
COVID-19 vaccines	Photophobia
Comirnaty and Vaxzevria	Haemophagocytic lymphohistiocytosis (HLH)
Comirnaty and Vaxzevria	Arterial thromboses
Vaxzevria	Acute disseminated encephalomyelitis, encephalitis, and encephalopathy
Comirnaty	Spontaneous abortion
Vaxzevria	Acute aseptic arthritis
Vaxzevria	Guillain-Barre syndrome
Vaxzevria	Retinal vessel thrombosis and occlusion
Spikevax	Myocarditis/Pericarditis
Comirnaty	Spontaneous abortion
Vaxzevria	Giant cell arteritis
Comirnaty and Vaxzevria	Erythema multiforme
Comirnaty	Paraesthesia and Hypoesthesia
Comirnaty	Glomerulonephritis and Nephrotic syndrome
Vaxzevria	Hearing disorders including deafness and tinnitus
Vaxzevria	Acute macular neuroretinopathy, AMOR and PAMM
COVID-19 vaccines	Multisystem inflammatory syndrome in children (MIS-C)
COVID-19 vaccines	Menstrual disorder
COVID-19 vaccines	Multisystem inflammatory syndrome in adults (MIS-A)
Vaxzevria	Transverse myelitis
Comirnaty	Multisystem inflammatory syndrome in children (MIS-C)
Spikevax	Urticaria
COVID-19 vaccines	Multisystem inflammatory syndrome in adults (MIS-A)
Nuvaxovid	Tinnitus, hypoacusis and ear discomfort
Nuvaxovid	Myocarditis/Pericarditis
Nuvaxovid	Hypoaesthesia and paraesthesia
COVID-19 vaccines	Multisystem inflammatory syndrome in children (MIS-C)
COVID-19 vaccines	Multisystem inflammatory syndrome in adults (MIS-A) and Multisystem inflammatory syndrome in children (MIS-C)
Vaxzevria	Myocardial infarct
Vaxzevria	Acute disseminated encephalomyelitis (ADEM)
COVID-19 vaccines	Uveitis
Comirnaty	Appendicitis

SQ25-001584 - ATTACHMENT A

Spikevax	Capillary leak syndrome
COVID-19 vaccines	Vulvovaginal ulcerations (Lipschutz ulcers) in adolescent girls
COVID-19 vaccines	Migraine
Comirnaty and Vaxzevria	Type I Diabetes Mellitus
Comirnaty	Anti-glomerular basement membrane disease
Comirnaty	Neuralgic amyotrophy
Comirnaty	Neutropenia
Comirnaty	Central serous chorioretinopathy
COVID-19 vaccines	Alopecia
Comirnaty	Sjogren's syndrome
Comirnaty	Cardiomyopathy
COVID-19 vaccines	Mastitis (Including breast swelling)
COVID-19 vaccines	Encephalitis
Comirnaty and Spikevax	Myocarditis/pericarditis
Comirnaty	Guillain-Barre syndrome
COVID-19 vaccines	Polymyalgia rheumatica
COVID-19 vaccines	Arterial dissection
COVID-19 vaccines	Dosing errors in children
Comirnaty and Spikevax	Heavy menstrual bleeding
COVID-19 vaccines	Haemophagocytic lymphohistiocytosis (HLH)
COVID-19 vaccines	Shoulder injury after vaccination administration (SIRVA)
COVID-19 vaccines	Chest pain
Comirnaty	Postural orthostatic tachycardia syndrome (POTS)
COVID-19 vaccines	Acquired haemophilia A
COVID-19 vaccines	Interstitial lung disease
COVID-19 vaccines	Myasthenia gravis
COVID-19 vaccines	Deafness/ hearing loss
Comirnaty	IgA nephropathy
Comirnaty	Glomerulonephritis minimal lesion
COVID-19 vaccines	Menstrual cycle irregularities
Comirnaty	Thyroid disorders
Vaxzevria	Superficial vein thrombosis
COVID-19 vaccines	Eosinophilic myocarditis
COVID-19 vaccines	Retinal vascular occlusion (RVO + RAO) and macular neuroretinopathy
Comirnaty	Acute disseminated encephalomyelitis
COVID-19 vaccines	Atrial fibrillation
Comirnaty	Multiple sclerosis
COVID-19 vaccines	Chest pain
COVID-19 vaccines	Type I Diabetes Mellitus
Nuvaxovid	Myocarditis/pericarditis
Comirnaty and Spikevax	Splanchnic venous thrombosis
Comirnaty and Spikevax	Thrombocytopenia and ITP
Comirnaty and Spikevax	Pulmonary embolism
Spikevax	Acute disseminated encephalomyelitis
Comirnaty and Spikevax	Generalised seizures
Comirnaty	Cerebral venous sinus thrombosis

SQ25-001584 - ATTACHMENT A

Comirnaty	Cerebral venous sinus thrombosis
Comirnaty and Spikevax	mRNA COVID-19 vaccines and post-menopausal haemorrhage / bleeding
COVID-19 vaccines	Rubbery clots (fibroelastic vascular casts)
COVID-19 vaccines	Long-COVID
Spikevax	Cerebral venous sinus thrombosis
Spikevax	Chronic urticaria
Comirnaty and Spikevax	Autoimmune haemolytic anaemia
COVID-19 vaccines	Incorrect administration of multi dose vials of COVID-19 vaccines
COVID-19 vaccines	Incorrect administration of multi dose vials of COVID-19 vaccines
COVID-19 vaccines	Bell's palsy
COVID-19 vaccines	Adverse events with a fatal outcome
Comirnaty and Spikevax	Cerebral venous sinus thrombosis
Comirnaty	Mast cell activation syndrome (MCAS)
Comirnaty	Myelin oligodendrocyte glycoprotein antibody associated disease (MOGAD)
Comirnaty	Neutropenia
Comirnaty	NUT carcinoma

Senate Committee: Community Affairs Committee

QUESTION ON NOTICE

Supplementary Budget Estimates 2025-2026

Outcome:

PDR Number: SQ25-001584

Question Subject: Covid-19 Vaccine Safety Monitoring Plan

Type of Question: Written

Senator: Alex Antic

Question:

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20. Could the TGA provide documentary evidence demonstrating how each of the five key strategies of the COVID-19 Vaccine Safety Monitoring Plan was put into in practice and the documented outputs of that plan including but not limited to:-
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 - e. close collaboration and coordination of effort with other vaccine safety stakeholder groups.

Documents evidencing the TGA's implementation of the Plan's five key strategies including the TGA's implementation of each of the strategies, objectives and outputs referenced in the key strategies of the Plan as follows:

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- b. Enhanced safety signal detection and investigation
- c. Actions in response to safety concerns
- d. Communications
- e. Collaborations

Any revisions of the Plan.

Answer:

17. The Therapeutic Goods Administration (TGA) has now investigated 150 potential safety signals in association with COVID-19 vaccines. These are listed in the table at Attachment A.

18. Since May 2021, the TGA has completed 57 regulatory actions to include new safety information in COVID-19 vaccine Product Information (PI) documents and, where relevant, corresponding Consumer Medicine Information (CMI) documents. These safety updates were initiated after the completion of TGA safety investigations.

Broken down by COVID-19 vaccine product, there have been 22 regulatory actions related to Comirnaty, 14 to Vaxzevria, 14 to Spikevax, and 7 to Nuvaxovid. There have been no regulatory actions related to the Janssen COVID-19 vaccine.

19. The five key elements articulated in the COVID-19 Vaccine Safety Monitoring Plan describe the internationally agreed practice of pharmacovigilance that is the core work of the TGA's Pharmacovigilance Branch. For COVID-19 vaccines, the TGA's already robust pharmacovigilance processes were dramatically enhanced, making it the most intensive safety monitoring of therapeutic goods ever conducted in Australia.

There is ample information in the public domain providing evidence of compliance with the plan and underscoring our commitment to transparency as a regulator. The TGA has released documents relating to COVID-19 vaccines in response to over 70 Freedom of Information requests, which are publicly available on the TGA's website: www.tga.gov.au/about-tga/freedom-information-foi.

20. Demonstrated examples of the how each of the five key strategies / objectives of the COVID-19 Vaccine Safety Monitoring Plan was put into in practice include:

a. Enhanced reporting of adverse events following immunisation - Timely collection and management of reports of COVID-19 vaccine adverse events following immunisation.

- During the COVID-19 vaccine roll out, entry of adverse events reported for COVID-19 vaccines into the TGA internal database was prioritised.
- The TGA's Database of Adverse Events Notification (DAEN) and the COVID-19 vaccine weekly safety reports, both publicly available, capture information on adverse events following COVID-19 vaccination reported to the TGA.

- Adverse event reports submitted to the TGA are assigned a case number, coded by the TGA with the relevant terms from the Medical Dictionary for Regulatory Activities, and entered into the TGA's internal database. De-identified information from adverse event reports is uploaded to the publicly accessible DAEN.
- As of 19 August 2021, the TGA reduced the time between adverse event reports being accepted into its database and published on the DAEN from 90 days to 14 days. This allowed information on reports for vaccines to be made publicly available more quickly, to meet the strong public interest relating to COVID-19 vaccinations.

b. Enhanced safety signal detection and investigation - Timely detection and investigation of COVID-19 vaccine safety signals

- Adverse event reporting data undergoes analysis to detect patterns of events that indicate a possible safety issue, or safety signals. The TGA conducts regular statistical analyses of these data to detect signals, in addition to closely monitoring the occurrence of 'adverse events of special interest'.
- Investigation of safety signals may involve activities more detailed analysis of adverse event report data. Investigation may also involve the review of published literature, information from medicines regulators in other countries, and safety data from international use of the vaccine provided.
- The TGA continues to investigate safety signals associated with COVID-19 vaccines.

c. Actions in response to safety concerns - Timely action to address any COVID-19 vaccine safety concerns

- Since May 2021, the TGA has completed 57 regulatory actions to include new safety information for COVID-19 vaccines in PI (for health professionals) and CMI documents (Attachment B).

d. Communications - Timely communications to inform the public of emerging COVID-19 vaccine safety information and to support public confidence in vaccines

- The TGA consistently and reliably published safety information about COVID-19 vaccines to the public in the COVID-19 vaccine safety report. This included summaries of the latest safety information, including outcomes of significant investigations, and information about regulatory actions taken as a result of the TGA's safety monitoring. The TGA published 104 reports over a period of more than 2 years. These can be accessed at: www.tga.gov.au/periodic/covid-19-vaccine-weekly-safety-report.
- To support public understanding and confidence, the TGA also issued media statements and safety updates to alert Australians to serious safety concerns, provide reassurance, and correct misinformation as required.
- The TGA also publishes safety updates on the TGA's Safety updates webpage at <https://www.tga.gov.au/news/safety-updates>. An example is the update in May 2024, which reported on the addition of migraine to the PI for Spikevax XXB.1.5

e. Collaborations - Close collaboration and coordination of effort with other vaccine safety stakeholder groups.

- The TGA continues to work closely with states and territories, other Australia government agencies, overseas regulators and key expert bodies, in order to ensure a coordinated approach to COVID-19 vaccine safety. Examples of these collaborations include:
 - alongside the UK's Medicines and Healthcare products Regulatory Agency, co-chairing regular meetings of international medicines regulators to share information about COVID-19 vaccine safety
 - holding regular meetings with state and territory jurisdictional immunisation coordinators to discuss emerging safety information and communications
 - providing regular briefings to the Australian Technical Advisory Group on Immunisation COVID-19 working groups to inform aspects of the vaccine roll-out
 - working with the National Centre for Immunisation Research and Surveillance to share information and characterise significant vaccine safety issues
 - seeking expert advice from the Advisory Committee for Vaccines on COVID-19 vaccine monitoring activities and safety issues.

Table 1 – List of COVID-19 vaccine safety updates

Active Name	PI Safety Updates
Comirnaty ((BNT162b2 [mRNA]) COVID-19 Vaccine	Severe allergic reactions
Comirnaty ((BNT162b2 [mRNA]) COVID-19 Vaccine	Anaphylaxis
Comirnaty ((BNT162b2 [mRNA]) COVID-19 Vaccine	Other hypersensitivity reactions (e.g. rash, pruritus, urticaria, angioedema)
Comirnaty ((BNT162b2 [mRNA]) COVID-19 Vaccine	Diarrhoea
Comirnaty ((BNT162b2 [mRNA]) COVID-19 Vaccine	Vomiting
Comirnaty ((BNT162b2 [mRNA]) COVID-19 Vaccine	Pain in extremity (vaccinated arm)
Comirnaty ((BNT162b2 [mRNA]) COVID-19 Vaccine	Myocarditis
Comirnaty ((BNT162b2 [mRNA]) COVID-19 Vaccine	Pericarditis
Comirnaty ((BNT162b2 [mRNA]) COVID-19 Vaccine	Syncope
Comirnaty ((BNT162b2 [mRNA]) COVID-19 Vaccine	Erythema multiforme
Comirnaty ((BNT162b2 [mRNA]) COVID-19 Vaccine	Paraesthesia
Comirnaty ((BNT162b2 [mRNA]) COVID-19 Vaccine	Hypoesthesia
Comirnaty ((BNT162b2 [mRNA]) COVID-19 Vaccine	Clarification of myocarditis warning including fatal cases and atypical presentations
Comirnaty ((BNT162b2 [mRNA]) COVID-19 Vaccine	Clarification of pericarditis warning including fatal cases and atypical presentations
Comirnaty ((BNT162b2 [mRNA]) COVID-19 Vaccine	Non-sexually acquired genital ulceration
Comirnaty ((BNT162b2 [mRNA]) COVID-19 Vaccine	Heavy menstrual bleeding
Comirnaty ((BNT162b2 [mRNA]) COVID-19 Vaccine	Mastitis
Comirnaty ((BNT162b2 [mRNA]) COVID-19 Vaccine	Breast Swelling
Comirnaty ((BNT162b2 [mRNA]) COVID-19 Vaccine – strain variant tozinameran	Migraine
Comirnaty ((BNT162b2 [mRNA]) COVID-19 Vaccine – strain variant raxtozinameran	Migraine
Comirnaty ((BNT162b2 [mRNA]) COVID-19 Vaccine – strain variant riltozinameran/tozinameran	Migraine

Comirnaty ((BNT162b2 [mRNA]) COVID-19 Vaccine – strain variant tozinameran/famtozinameran	Migraine
VAXZEVRIA (previously COVID-19 Vaccine AstraZeneca) ChAdOx1-S	Addition of thrombosis with thrombocytopenia syndrome (TTS)
VAXZEVRIA (previously COVID-19 Vaccine AstraZeneca) ChAdOx1-S	Addition of capillary leak syndrome (CLS)
VAXZEVRIA (previously COVID-19 Vaccine AstraZeneca) ChAdOx1-S	Addition of Guillain-Barre Syndrome (GBS)
VAXZEVRIA (previously COVID-19 Vaccine AstraZeneca) ChAdOx1-S	Anxiety-related reactions (includes syncope)
VAXZEVRIA (previously COVID-19 Vaccine AstraZeneca) ChAdOx1-S	Addition of immune thrombocytopenic purpura (ITP)
VAXZEVRIA (previously COVID-19 Vaccine AstraZeneca) ChAdOx1-S	Addition of VT events without thrombocytopenia in section 4.4; CVST without thrombocytopenia in section 4.8
VAXZEVRIA (previously COVID-19 Vaccine AstraZeneca) ChAdOx1-S	TTS reporting rates lower with 2nd dose
VAXZEVRIA (previously COVID-19 Vaccine AstraZeneca) ChAdOx1-S	Removed causality not established for GBS
VAXZEVRIA (previously COVID-19 Vaccine AstraZeneca) ChAdOx1-S	Transverse myelitis
VAXZEVRIA (previously COVID-19 Vaccine AstraZeneca) ChAdOx1-S	Hypoesthesia
VAXZEVRIA (previously COVID-19 Vaccine AstraZeneca) ChAdOx1-S	Paraesthesia
VAXZEVRIA (previously COVID-19 Vaccine AstraZeneca) ChAdOx1-S	Immune thrombocytopenia
VAXZEVRIA (previously COVID-19 Vaccine AstraZeneca) ChAdOx1-S	Tinnitus
VAXZEVRIA (previously COVID-19 Vaccine AstraZeneca) ChAdOx1-S	Acute disseminated encephalomyelitis and cutaneous vasculitis
SPIKEVAX (elasomeran) COVID-19 VACCINE	Strengthening of warning on myocarditis and pericarditis
SPIKEVAX (elasomeran) COVID-19 VACCINE	Hypersensitivity reactions (e.g. erythema multiforme)
SPIKEVAX (elasomeran) COVID-19 VACCINE	Paraesthesia and hypoesthesia
SPIKEVAX (elasomeran) COVID-19 VACCINE	Capillary leak syndrome (CLS)
SPIKEVAX (elasomeran) COVID-19 VACCINE	Clarification of myocarditis warning including fatal cases and atypical presentations
SPIKEVAX (elasomeran) COVID-19 VACCINE	Clarification of pericarditis warning including fatal cases and atypical presentations
SPIKEVAX (elasomeran) COVID-19 VACCINE	Heavy menstrual bleeding
SPIKEVAX (elasomeran) COVID-19 VACCINE	Migraine

SPIKEVAX (elasomeran) COVID-19 VACCINE	Mastitis
SPIKEVAX (elasomeran) COVID-19 VACCINE	Breast Swelling
SPIKEVAX (elasomeran) COVID-19 VACCINE- strain variant SPIKEVAX BIVALENT ORIGINAL OMICRON BA.4-5 (elasomeran/davesomeran)	Chronic urticaria
SPIKEVAX (elasomeran) COVID-19 VACCINE- strain variant SPIKEVAX JN.1	Chronic urticaria
SPIKEVAX (elasomeran) COVID-19 VACCINE- strain variant SPIKEVAX XBB.1.5 COVID-19 VACCINE	Chronic urticaria
SPIKEVAX (elasomeran) COVID-19 VACCINE- strain variant SPIKEVAX (elasomeran)	Chronic urticaria
NUVAXOVID (SARS-CoV-2rS [NVX-CoV2373]) COVID-19 VACCINE	Anaphylaxis
NUVAXOVID (SARS-CoV-2rS [NVX-CoV2373]) COVID-19 VACCINE	Pericarditis
NUVAXOVID (SARS-CoV-2rS [NVX-CoV2373]) COVID-19 VACCINE	Hypoaesthesia
NUVAXOVID (SARS-CoV-2rS [NVX-CoV2373]) COVID-19 VACCINE	Paraesthesia
NUVAXOVID (SARS-CoV-2rS [NVX-CoV2373]) COVID-19 VACCINE	Myocarditis
NUVAXOVID (SARS-CoV-2rS [NVX-CoV2373]) COVID-19 VACCINE	Tinnitus
NUVAXOVID (SARS-CoV-2rS [NVX-CoV2373]) COVID-19 VACCINE	Migraine